

ORIGINAL ARTICLE

Efficacy, safety and predictive biomarker of third-generation tyrosine kinase inhibitors with azacitidine in myeloid blast phase of chronic myeloid leukemia

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Abstract

Objective: To evaluate the efficacy, safety, and predictive biomarker of a third-generation tyrosine kinase inhibitor (3G-TKI; ponatinib or olverembatinib) combined with azacitidine in chronic myeloid leukemia (CML) in myeloid blast phase.

Methods: We conducted a single-center, prospective study combining 3G-TKI with azacitidine in 28-day cycles. The primary end point was a major hematologic response (MaHR) by cycle 2. The trial is registered in Chinese Clinical Trial Registry (ChiCTR2200055887)

Results: In total, 37 patients were studied. The median follow-up was 30 months (interquartile range, 24–40 months). Twenty-five patients achieved a MaHR by cycle 2, 30 returned to chronic phase. Ten patients underwent transplantation. The patients who underwent transplantation had higher 3-year probability of survival compared with nontransplanted patients (50%; [95% confidence interval (CI), 9%–37%] versus 18% [95% CI, 3%–33%]; $p = .01$). The regimen was well tolerated. In adjusted logistic/Cox regression analyses, *KRAS* mutation was significantly associated with a lower MaHR rate (odds ratio, 0.1; 95% CI, 0–0.8; $p = .03$), worse progression-free survival (PFS; hazard ratio [HR], 3.1; 95% CI, 1.1–8.6; $p = .04$), and worse survival (HR, 8.2; 95% CI, 2.5–26.8; $p < .001$); *PTPN11* mutation was associated with worse PFS (HR, 5.1; 95% CI, 1.2–22.2; $p = 0.03$) and worse survival (HR, 9.6; 95% CI, 2.2–41.5; $p = .002$); and increasing numbers of non-*ABL1* mutations were associated with worse PFS (HR, 1.2;

The first two authors contributed equally to this article.

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95% CI, 1.0–1.3; $p = .04$). Transcriptomic analysis revealed that patients who did not achieve a MaHR experienced activation of cancer-, metabolism-, oxidative phosphorylation-related pathways. The *KRAS* signaling pathway was significantly activated in patients who lost MaHR during treatment.

Conclusions: 3G-TKI with azacitidine is an effective and safe therapy providing more chance to receive a transplantation for CML in myeloid blast phase. Potential biomarkers associated with outcomes were identified.

KEYWORDS

azacitidine, chronic myeloid leukemia, myeloid blast phase, olverembatinib, ponatinib

INTRODUCTION

Most people with chronic myeloid leukemia (CML) in chronic phase have a normal life expectancy in the era of tyrosine kinase inhibitors (TKIs).¹ However, nearly 5% of people with CML in chronic phase experience transformation to blast phase.² The optimal treatment strategy remains controversial.³ Most interventions attempt to re-establish chronic phase, sometimes followed by hematopoietic cell transplantation.⁴ The European LeukemiaNet (ELN) and National Comprehensive Cancer Network (NCCN) guidelines recommend dasatinib or ponatinib with intensive acute myeloid leukemia-like regimens.^{1,5} In the MATCHPOINT study, ponatinib combined with fludarabine, cytarabine, idarubicin, and granulocyte colony-stimulating factor lead to a hematologic response of rate 69% in patients with CML in blast phase, whereas the death rate from treatment-related events was 18%.⁶ Previous studies demonstrated TKIs combined with hypomethylating agent had hematological response of 30–81% and were well-tolerated in those with CML in myeloid blast phase.^{7–9} Imatinib and second-generation TKIs were the most commonly used in these studies. Recently, Short et al.¹⁰ and Philippe et al.¹¹ reported the promising efficacy of ponatinib and azacitidine with or without venetoclax. However, data on third-generation TKIs (3G-TKI, such as ponatinib and olverembatinib) in this setting remain limited. Moreover, the absence of prognostic biomarkers highlights the further exploration of the approaches for CML in myeloid blast phase.

Aggregated data demonstrated that non-*ABL1* mutations, such as *RUNX1*, *ASXL1*, *KRAS*, and *PTPN11* mutations, were frequently identified in patients with CML in blast phase.^{12,13} Ochi et al. reported that patients with CML in blast phase who carried *ASXL1* mutations, complex copy number alterations, and trisomy 21 had shorter survival.¹³ Previous studies indicated that *KRAS* or *PTPN11* mutations were associated with TKI resistance in patients with chronic phase CML.^{14,15}

Therefore, we designed and conducted a single-center, prospective study to evaluate the safety and efficacy of 3G-TKIs with

azacitidine in patients with CML in myeloid blast phase and to identify clinical and biologic covariates correlated with outcomes.

MATERIALS AND METHODS

Patients

We enrolled 37 consecutive adult patients with CML in blast phase according to World Health Organization 2016 criteria. Detailed inclusion and exclusion criteria are listed in Table S1.

Study design

Patients received a 3G-TKI at an initial dose of ponatinib 45 mg once daily or olverembatinib 30 mg every other day continuously combined with azacitidine (75 mg/m² once daily for 7 days) in 28-day cycles from June 2021 to June 2023. The choice of 3G-TKI was based on availability. Doses were adjusted based on responses and adverse events (AEs); details are provided shown in Table S2.

The primary end point was major hematologic response (MaHR) after 2 cycles. Secondary end points were safety, complete hematologic response (CHR), major cytogenetic response (MCyR), complete cytogenetic response (CCyR), major molecular response (MMR), progression-free survival (PFS), and survival (defined below). Eligible patients who returned to chronic phase underwent allo-transplantation. Donor selection included human leukocyte antigen (HLA)-matched sibling donors and haploidentical or HLA unrelated donors. The consensus on indications, conditioning regimen, and donor selection of transplantation has been described in previous studies.^{16,17} The date of last follow-up was March 15, 2025. Genomic and transcriptomic analyses were used to identify biomarkers correlated with outcomes. The study was approved by the Ethics Committee of Peking University People's Hospital. All patients provided written informed consent. The study was registered in the

Chinese Clinical Trial Registry (ChiCTR2200055887; <https://www.chictr.org.cn/index.html>).

Diagnosis, definitions and monitoring

Myeloid blast phase was defined according to WHO 2016 criteria, including $\geq 20\%$ blasts in the blood or bone marrow.¹⁸ Hematologic responses included: (1) MaHR: either a CHR or no evidence of leukemia (NEL)¹⁹; (2) CHR: white blood cells (WBC) $< 10 \times 10^9/L$, absolute neutrophil count (ANC) $\geq 1 \times 10^9/L$, platelet concentration $\geq 100 \times 10^9/L$, no blasts or promyelocytes in blood or blasts in bone marrow $\leq 5\%$, blood myelocytes and metamyelocytes $< 5\%$, blood basophils $< 5\%$, and no extramedullary leukemia; (3) NEL: same as CHR criteria except for platelet concentration $\geq 20 \times 10^9/L$ and $< 100 \times 10^9/L$ and/or ANC $\geq 0.5 \times 10^9/L$ and $< 1 \times 10^9/L$; and (4) return to chronic phase: $< 10\%$ blasts in blood and bone marrow with no extramedullary leukemia.

Cytogenetic and molecular responses were defined using ELN recommendations, including: (1) MCyR: Philadelphia chromosome (Ph)-positive cells $\leq 35\%$ in ≥ 20 bone marrow metaphases; (2) CCyR: no Ph-positive cells; (3) MMR: International Scale BCR::ABL (BCR::ABL1^{IS}) $\leq 0.1\%$; (4) molecular response 4 (MR⁴): BCR::ABL1^{IS} $\leq 0.01\%$; and (5) molecular response 4.5 (MR^{4.5}): BCR::ABL1^{IS} $\leq 0.0032\%$. Additional cytogenetic abnormalities (ACAs) in Ph-positive cells were identified as previously described, and high-risk ACAs were defined according to the 2020 ELN recommendations, including trisomy 8, second Ph chromosome, i(17q), trisomy 19, -7/del(7q), 11q23, 3q26.2 aberrations, and complex aberrant karyotype.^{1,20} All ACAs were independently reviewed by two cytogeneticists. When the interpretations were inconsistent, they were referred to a senior cytogeneticist for re-evaluation.

PFS was defined as the interval from therapy start to increasing WBC concentration in patients without MaHR (WBC concentration $> 20 \times 10^9/L$ on two observations after one cycle), loss of MaHR or MCyR but still in chronic phase, progression to accelerated phase or blast phase after return to chronic phase, death from any cause, withdrawal of consent or discontinuation from study (whichever occurred first), or censored at the last follow-up.²¹ Survival was defined as the interval from therapy start to death from any cause, censored at last follow-up, or withdrawal of consent.

Clinic visits were at least once a week during the first one or two cycles and once every 2 or 4 weeks thereafter. Complete blood counts with differentials, as well as renal, liver function, and electrolyte levels, were assessed at baseline and at each subsequent visit. Electrocardiograms were obtained at baseline and as clinically indicated.

Hematologic response, immune phenotype, cytogenetic and molecular responses were evaluated at baseline, after the first two cycles, and every three cycles thereafter or as clinically indicated. Tests for ABL1 mutation by Sanger sequencing and targeted DNA sequencing of blood or bone marrow samples were done at baseline,

at the end of the first and second cycles, and when clinically indicated.

Bone marrow cytogenetic analyses used G-banding. Blood BCR::ABL1 transcript levels were analyzed by quantitative real-time polymerase chain reaction with an ABL1 control and were converted to the BCR::ABL1^{IS} value using our laboratory-specific conversion factor of 0.65 validated at the Institute of Medical and Veterinary Science International Reference Laboratory when the value (IS) was $< 10\%$.²²

Safety

Hematologic and nonhematologic AEs were scored according to the National Cancer Institute's Common Terminology Criteria for Adverse Events, version 5.0.

Targeted DNA sequencing

Targeted DNA sequencing was performed using the Illumina platform (Illumina Inc.; see *Targeted DNA sequencing* in the Supporting Methods, *Targeted DNA sequencing*; Table S3).

RNA sequencing and enrichment analyses

RNA sequencing was performed by Novogene Company Ltd. (see Supporting Methods, *RNA extraction, library preparation, and sequencing*). Gene set variation analysis (GSVA) and gene set enrichment analysis (GSEA) were conducted to evaluate pathway enrichment (see Supporting Methods).^{23,24}

Statistical analyses

Descriptive statistics are used to summarize covariates. Categorical covariates are reported as percentages and counts. Continuous variables are reported as medians and ranges or interquartile ranges (IQR). The Pearson χ^2 test was used to analyze categorical covariates. Student *t*-tests or Mann-Whitney *U* tests were used to analyze continuous covariates. Paired parameters were compared using the paired *t*-test. Comorbidity(ies) were defined according to diagnostic categories in the Charlson comorbidity index.²⁵ PFS and survival were estimated using the Kaplan-Meier method with the log-rank test. An adjusted logistic regression model was used to identify covariates associated with MaHR, adjusted Cox regression models, PFS, and survival. Data were adjusted for clinical covariates that were associated with MaHR, PFS, and/or survival in univariable analyses. A two-sided *p* value $< .05$ was considered significant. R version 4.0.2 (R Core Team) and GraphPad Prism 8 (GraphPad Software Inc.) were used for analyses and graphing.

RESULTS

Patients

Data from 75 consecutive patients were assessed for eligibility from July, 2021 to July, 2023; of these, 38 patients were excluded because of lymphoid blast phase ($n = 30$), cost ($n = 4$), personal preference ($n = 2$), isolated extramedullary leukemia ($n = 1$), or heart failure ($n = 1$). The remaining 37 consecutive patients with CML in myeloid blast phase were studied (Figure 1). Patient covariates are summarized in Table 1. Twenty patients (54%) were male. Median age was 50 years (range, 23–72 years; IQR, 33–54 years). Twenty-nine patients (78%) transformed from chronic phase, six (16%) transformed from accelerated phase, and two (5%) presented with de novo myeloid blast phase. The median interval from diagnosis of CML to myeloid blast phase was 31 months (IQR, 8–91 months). Twenty patients (54%) had received three or more prior TKIs. Ten patients (27%) had previously failed on ponatinib, olverembatinib, and/or TGRX-678 monotherapy in chronic or accelerated phase before the study treatment: five of them continued their previous 3G-TKI (ponatinib, $n = 4$; olverembatinib, $n = 1$), and five switched from TGRX-678 to ponatinib ($n = 1$) or olverembatinib ($n = 4$) combined with azacitidine. Twenty-six patients (70%) achieved a best response of CHR, five (14%) achieved a MCyR, 2 (5%) achieved a MMR, and two (5%) achieved a MR^{4,5} before the study treatment. The median interval from diagnosis of myeloid blast phase to study entry was 0.3

months (IQR, 0.1–1.0 month). Sixteen (43%) and 21 (57%) patients received ponatinib and olverembatinib, respectively (Table 1; see Table S4).

Cytogenetics

Twenty-nine patients (78%) had ACAs in Ph-positive cells at baseline, 26 of which were high-risk ACAs. Complex aberrant karyotype ($n = 12$; 32%) was the most frequent high-risk ACA, followed by trisomy 8 ($n = 11$; 30%), second Ph chromosome ($n = 11$; 30%), trisomy 21 ($n = 5$; 14%), 3q26.2 aberrations ($n = 5$; 14%), $-7/\text{del}(7q)$ ($n = 4$, 11%), isochromosome (17q) ($n = 3$; 8%), trisomy 19 ($n = 2$; 5%), and 11q23 ($n = 2$; 5%; Figure 2A,B).

Genomics

ABL1 mutations were detected in 18 patients (49%), with a median number of two mutations (IQR, one to three mutations) per patient. The most frequent *ABL1* mutations were observed in T315I ($n = 10$; 27%), followed by E255K ($n = 8$; 22%), F359I ($n = 3$; 8%), and F359V ($n = 3$; 8%; Figure 2A–C). Non-*ABL1* somatic mutations were detected in 33 patients (89%), with a median number of three mutations (IQR, 2–5 mutations) per patient. The most frequent mutations was observed in ASXL1 ($n = 11$; 30%), followed by GATA2 ($n = 10$; 27%),

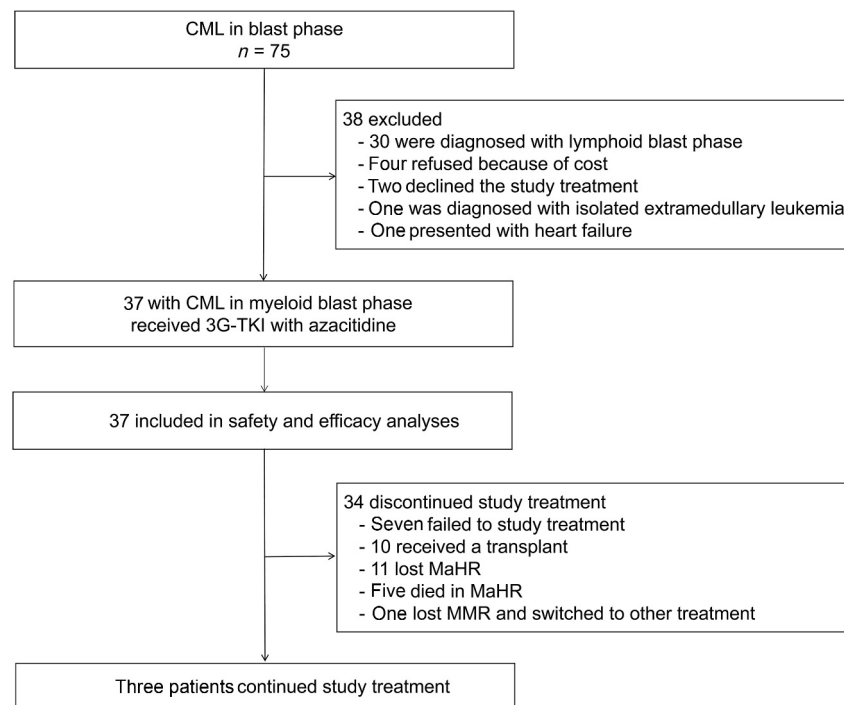


FIGURE 1 Study flow diagram. 3G-TKI indicates third-generation tyrosine kinase inhibitor; CML, chronic myeloid leukemia; MaHR, major hematologic response; MMR, major molecular response.

TABLE 1 Patient characteristics, $N = 37$.

Variable	No. (%)
Age: Median [IQR], years	50 [33–54]
Male	20 (54.0)
Comorbidity (ies)	11 (30.0)
Disease phase at diagnosis	
CP	29 (78.0)
AP	6 (16.0)
CML-MBP	2 (5.0)
Interval from diagnosis of CML to CML-MBP: Median [IQR], months	31 [8–91]
Prior TKI therapy lines before study treatment	
1	8 (22.0)
2	8 (22.0)
≥3	20 (54.0)
None	1 (3.0)
Prior TKI used	
Imatinib	29 (78.0)
Dasatinib	26 (70.0)
Nilotinib	19 (51.0)
Flumatinib	10 (27.0)
Ponatinib	3 (8.0)
Olverembatinib	3 (8.0)
TGRX-678	5 (14.0)
None	1 (3.0)
Best treatment response before CML-MBP	
≥CHR and < PCyR	26 (70.0)
≥PCyR and < MMR	5 (14.0)
≥MMR	4 (11.0)
NA	2 (5.0)
WBC: Median [range], ($\times 10^9/L$)	8 [0.4–313]
Hemoglobin: Median [range], g/L	82 [41–150]
Platelet count: Median [range], $\times 10^9/L$	77 [3–810]
Blast count: Median [range], %	40 [20–95]
Interval from CML-MBP to therapy start [IQR], months	0.2 [0.1–1.0]
3G-TKI used in study treatment	
Ponatinib	16 (43.0)
Olverembatinib	21 (57.0)

Abbreviations; 3G-TKI, third-generation tyrosine kinase inhibitor; AP, accelerated phase; CHR, complete hematologic response; CCyR, complete cytogenetic response; CML, chronic myeloid leukemia; CML-MBP, chronic myeloid leukemia in myeloid blast phase; CP, chronic phase; IQR, interquartile range; MMR, major molecular response; NA, not available; PCyR, partial cytogenetic response; TKI, tyrosine kinase inhibitor; WBC white blood cell.

RUNX1 ($n = 9$; 24%), *KRAS* ($n = 6$; 16%), *PTPN11* ($n = 3$; 8%), *TP53* ($n = 3$; 8%), *EP300* ($n = 3$; 8%), *DNMT3A* ($n = 3$; 8%), *PHF6* ($n = 3$; 8%), and *KMT2C* ($n = 3$; 8%).

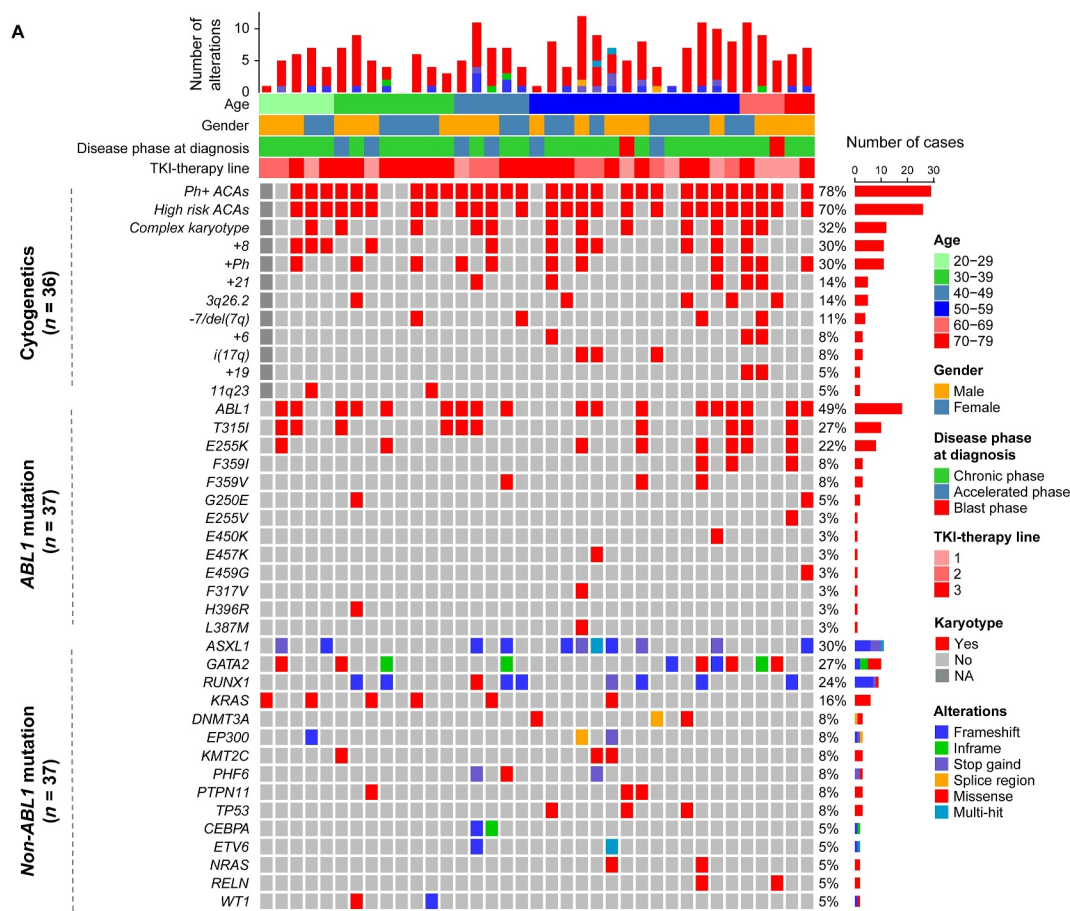
Efficacy

Ten patients (27%) received one cycle of study treatment, and 27 (73%) received two or more cycles. The median number of cycles was two (IQR, 1–4 cycles; Figure 3). 3G-TKI dose reductions and at least one dose interruption occurred in 26 patients (70%), with a median delay of 1 month (IQR, 1–2 months). Delays in azacitidine treatment occurred in 18 patients (49%), with a median delay of 1 month (IQR, 1–3 months). Twenty-three patients (62%) achieved a MaHR (CHR, $n = 7$; NEL, $n = 16$), and 30 (81%) returned to chronic phase after one cycle. Twenty-five patients (68%) achieved a MaHR (CHR, $n = 7$; NEL, $n = 18$), and 30 (81%) returned to chronic phase by two cycles. One of 25 patients (3%) who achieved a MaHR died of heart failure at 31 days, had no blasts in either bone marrow or blood, and had a *BCR::ABL1*¹⁵ transcript level of 1.2% detected before death. The median duration of MaHR was 4 months (IQR, 1–12 months) in those who did not undergo transplantation. During study treatment, 13 patients (35%) achieved MCyR, nine (24%) achieved CCyR, and six (16%) achieved MMR.

Among seven patients who did not return to chronic phase within two cycles, three switched to ponatinib with a 3 + 7 regimen, four switched to ponatinib or olverembatinib (two each) with azacitidine plus venetoclax, and all seven died with no response at a median of 6 months (IQR, 3–10 months).

Of the 30 responders (MaHR, $n = 25$; return to chronic phase, $n = 5$), ten underwent myeloablative haploidentical hematopoietic cell transplantation. The remaining 19 patients did not proceed to transplantation because of cost ($n = 11$), unfit status ($n = 5$), or personal preference ($n = 3$). Except one patient with no myeloid engraftment, nine patients received ponatinib ($n = 5$), olverembatinib ($n = 3$), or dasatinib ($n = 1$) at 3 months after transplantation. Two patients subsequently progressed to myeloid blast phase again at 4 and 5 months and died; three died of transplantation-related complications at 0.4, 2, and 5 months; and one experienced a molecular relapse at 8 months after transplantation and received a second allotransplant. At the last follow-up, five transplantation recipients were alive with undetectable *BCR::ABL1*¹⁵ transcripts at a median of 30 months (IQR, 24–37 months), including three who received olverembatinib ($n = 2$) or dasatinib ($n = 1$) and two who discontinued TKI therapy (Figure 3).

Twenty patients who did not undergo transplantation continued to receive 3G-TKI therapy with azacitidine. Eleven patients progressed to a second blast phase (myeloid, $n = 10$; lymphoid, $n = 1$) at a median of 8 months (IQR, 5–12 months); 5 patients died of coronavirus disease 2019, other infections, or heart failure in MaHR; one achieved MR^{4.5} at 4 months, lost MMR at 19 months, subsequently received study treatment plus venetoclax, and re-achieved MMR; and three patients still continued study treatment with CHR ($n = 2$).



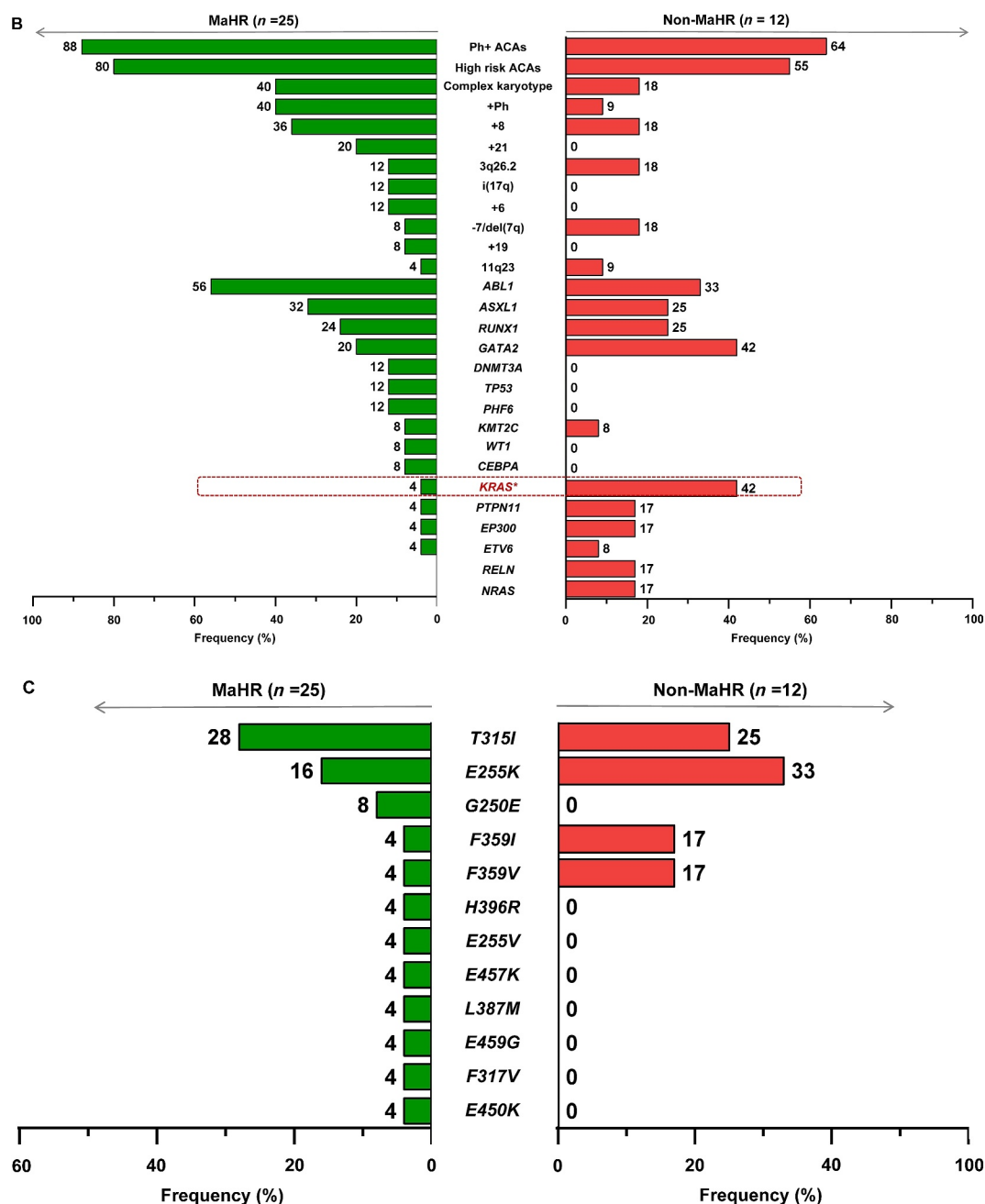


FIGURE 2 (Continued)

or MR⁴ ($n = 1$). Among 11 patients who progressed to a second blast phase, two who received ponatinib plus azacitidine switched to olverembatinib plus azacitidine with venetoclax, one who received olverembatinib plus azacitidine switched to olverembatinib plus idarubicin with cytarabine, and one who received ponatinib plus azacitidine switched to ponatinib plus homoharringtonine with decitabine. The remaining patients did not receive further treatment modifications, and all subsequently died shortly.

The median follow-up was 30 months (IQR, 24–40 months) in the patients who remained alive. The 3-year probabilities of PFS and survival were 19% (95% CI, 5%–32%) and 23% (95% CI, 9%–37%), respectively (Figure 3A,B), in all patients. The median PFS and survival were 5 months (95% CI, 2–7 months) and 9 months (95% CI, 6–12 months), respectively. Categorized by response, the 3-year probabilities of PFS (29% [95% CI, 11%–48%] vs. 0%; $p = .003$) and survival (39% [95% CI, 20%–58%] vs. 0%; $p = .005$) were superior for

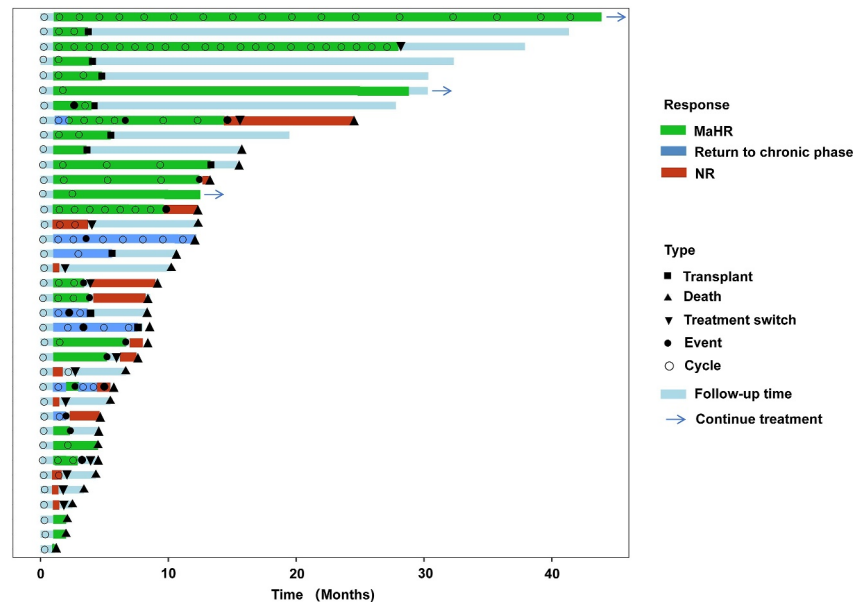


FIGURE 3 Swimming plot of response and outcomes. Events were defined as the loss of MaHR or major cytogenetic response or progression to accelerated or blast phase. MaHR indicates major hematologic response; NR, no response.

patients who achieved MaHR compared with those who did not, with a median PFS of 7 months (95% CI, 3–11 months) versus 2 months (95% CI, 1–3 months), respectively, and a median survival of 13 months (95% CI, 3–22 months) versus 7 months (95% CI, 2–11 months), respectively (Figure 4C,D).

Categorized by treatment, the 3-year probabilities of PFS (40% [95% CI, 10%–70%] vs. 11% [95% CI, 0%–25%]; $p = .04$) and survival (50% [95% CI, 19%–81%] vs. 18% [3%–33%]; $p = 0.01$) were superior for patients who underwent transplantation compared to those who did not (Figure 4E,F), with a median PFS of 11 months (95% CI, 2–19 months) versus 4 months (95% CI, 3–5 months), respectively, and a median survival of 16 months (95% CI, 11–21 months) versus 8 months (95% CI, 3–12 months), respectively.

Of the 10 patients who previously failed on ponatinib, olverembatinib, and/or TGRX-678 monotherapy in the chronic and/or accelerated phase, eight (80%) returned to chronic phase and achieved a MaHR during the study treatment, and four (40%) underwent transplantation. At a median follow-up of 30 months, the median PFS and survival were 4 months (95% CI, 0–9 months) and 12 months (95% CI, 9–16 months), respectively. Their MaHR rates ($p = .45$), PFS ($p = .58$), and survival ($p = .26$) were comparable to those in patients who had no prior 3G-TKI exposure.

Safety

The most common grade 3–4 AEs were thrombocytopenia ($n = 30$; 81%), neutropenia ($n = 26$; 70%), and anemia ($n = 25$; 68%; Table 2). No patient experienced grade 5 hematologic AEs. One patient died early because of heart failure at 31 days. Other non-hematologic AEs

were pulmonary infections ($n = 13$; 35%), hypocalcemia ($n = 12$; 32%) and fatigue ($n = 8$; 22%; Table 2). Five patients (14%) experienced grade 5 pulmonary infection (coronavirus disease 2019, $n = 2$; bacterial, $n = 1$), abdominal infection ($n = 1$), or heart failure ($n = 1$) at a median of 2 months (IQR, 2–8 months) and subsequently died. No patient experienced grade 4 nonhematologic AEs. Details of AEs are displayed in Table 2.

Identifying prognostic biomarker

Baseline clinical characteristics and molecular abnormalities were analyzed to explore the biomarkers associated with the study treatment response and outcomes. The results of adjusted logistic or Cox regression analyses are displayed in Table S5. Patients who had *KRAS* or *PTPN11* mutations had significantly worse PFS ($p = .02$ and $p = .006$, respectively) and survival ($p < .001$ and $p = .001$, respectively) compared with those who had wild-type *KRAS* or *PTPN11* (Figure 5A–D). The data were further adjusted by potentially relevant clinical covariates, including the interval from CML diagnosis to myeloid blast phase, WBC concentration, platelet concentration, and blast percentage. After adjusting these covariates, *KRAS* mutation was significantly associated with a lower MaHR rate (odds ratio, [95% CI, 0–0.8]; $p = .03$), worse PFS (hazard ratio [HR], 3.1 [95% CI, 1.1–8.6]; $p = .04$), and worse survival (HR, 8.2 [95% CI, 2.5–26.8]; $p < .001$); *PTPN11* mutation was associated with worse PFS (HR, 5.1 [95% CI, 1.2–22.2]; $p = .03$) and worse survival (HR, 9.6 [95% CI, 2.2–41.5]; $p = .002$); and increasing numbers of *non-ABL1* mutations were associated with worse PFS (HR, 1.2 [95% CI, 1.0–1.3]; $p = .04$; see Table S5).

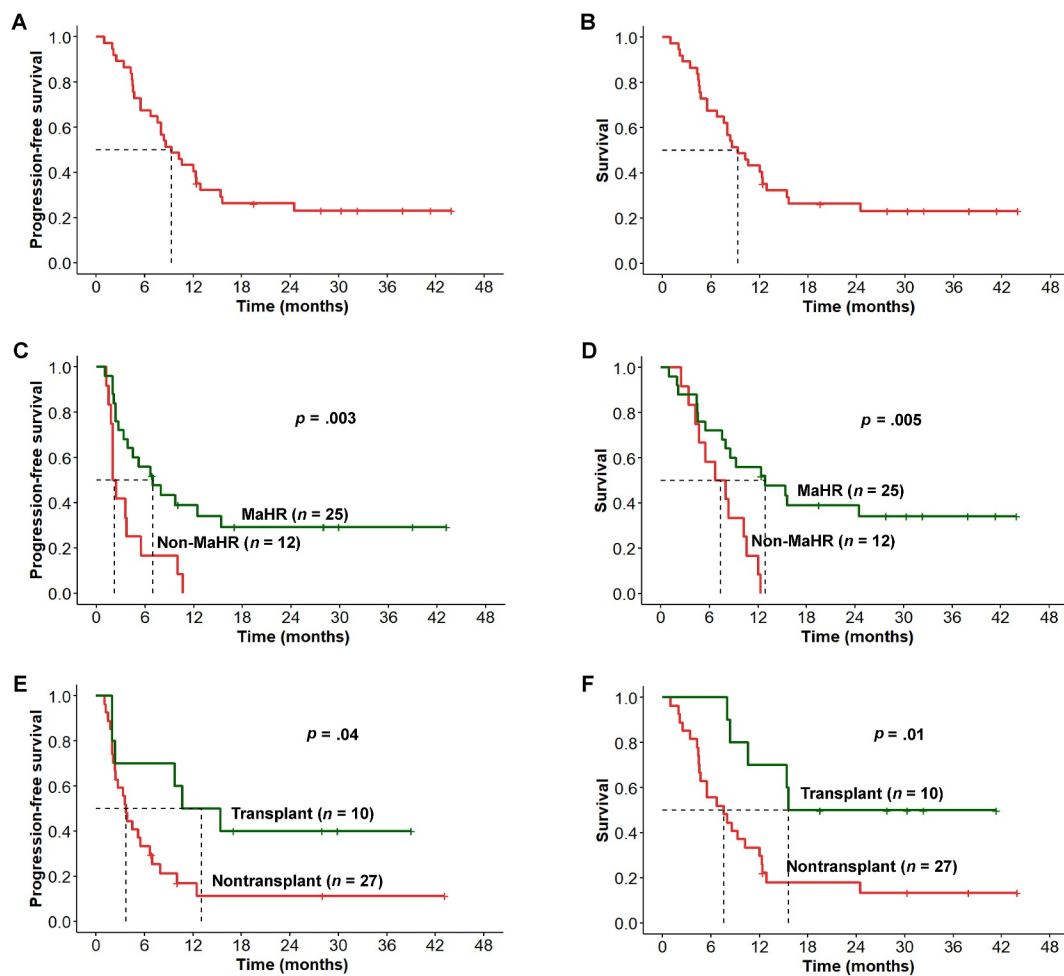


FIGURE 4 Progression-free survival and survival. (A,B) All patients; (C,D) comparison by MaHR status; and (E,F) comparison by transplantation status. MaHR indicates major hematologic response.

TABLE 2 Adverse events.

Event	No. (%)		
	Any grade	Grade 1-2	Grade 3
Hematologic			
Thrombocytopenia	32 (86.0)	2 (5.0)	30 (81.0)
Neutropenia	28 (76.0)	2 (5.0)	26 (70.0)
Anemia	26 (70.0)	1 (3.0)	25 (68.0)
Non-hematologic			
Febrile neutropenia	21 (57.0)	0 (0.0)	21 (57.0)
Pulmonary infection ^a	13 (35.0)	8 (22.0)	3 (8.0)
Hypocalcemia	12 (32.0)	12 (32.0)	0 (0.0)
Fatigue	8 (22.0)	8 (22.0)	0 (0.0)
Hyperbilirubinemia	7 (19.0)	5 (14.0)	2 (5.0)
Elevated alanine aminotransferase	6 (16.0)	4 (11.0)	2 (5.0)
Pleural effusion	5 (14.0)	5 (14.0)	0 (0.0)
Nausea and/or vomiting	5 (14.0)	5 (14.0)	0 (0.0)

(Continues)

TABLE 2 (Continued)

Event	No. (%)		
	Any grade	Grade 1-2	Grade 3
Pancreatitis	2 (5.0)	0 (0.0)	2 (5.0)
Atrial fibrillation	2 (5.0)	2 (5.0)	0 (0.0)
Elevated uric acid	2 (5.0)	2 (5.0)	0 (0.0)
Abdominal infection ^b	1 (3.0)	0 (0.0)	0 (0.0)
Cholecystitis	1 (3.0)	0 (0.0)	1 (3.0)
Heart failure ^c	1 (3.0)	0 (0.0)	0 (0.0)
Acute kidney injury	1 (3.0)	0 (0.0)	1 (3.0)
Epistaxis	1 (3.0)	1 (3.0)	0 (0.0)
Blood in phlegm	1 (3.0)	1 (3.0)	0 (0.0)
Gingival bleeding	1 (3.0)	1 (3.0)	0 (0.0)

^aThree patients developed grade 5 pulmonary infection.

^bOne patient developed grade 5 abdominal infection.

^cOne patient developed grade 5 heart failure.

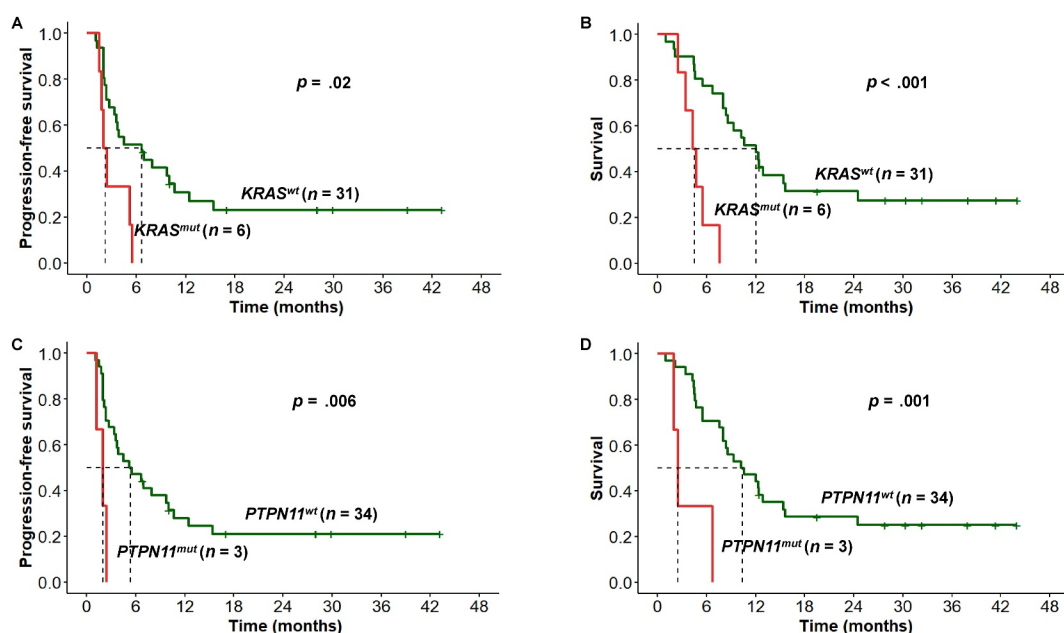


FIGURE 5 Impact of *KRAS* and *PTPN11* mutations on outcomes. (A,B) Impact of *KRAS* mutation on progression-free survival and survival. (C,D) Impact of *PTPN11* mutation on progression-free survival and survival. mut, indicates mutation; PB, peripheral blood; wt, wild type.

Transcriptomic analysis reveals pathway activation associated with treatment response

RNA sequencing was performed on samples from 37 patients for transcriptome-wide differential expression analysis (one patient was excluded because of early death). GSEA analyses revealed that the 12 patients who did not achieve MaHR had significantly higher activation of cancer-related pathways, such as the *TGF-β* signaling, mitotic

spindle, and oxidative phosphorylation pathways, compared with the 24 patients who achieved MaHR (Figure 6A).

Subsequently, we analyzed data from seven patients who did not achieve MaHR during study treatment. GSEA analyses revealed marked activation of cancer-related pathways (*MYC* targets v1, *MYC* targets v2, *mTORC1* signaling, *TGF-β* signaling, unfolded protein response, DNA repair, G2M checkpoint, and *E2F* targets pathways) and metabolism-related pathways (glycolysis, adipogenesis, fatty acid

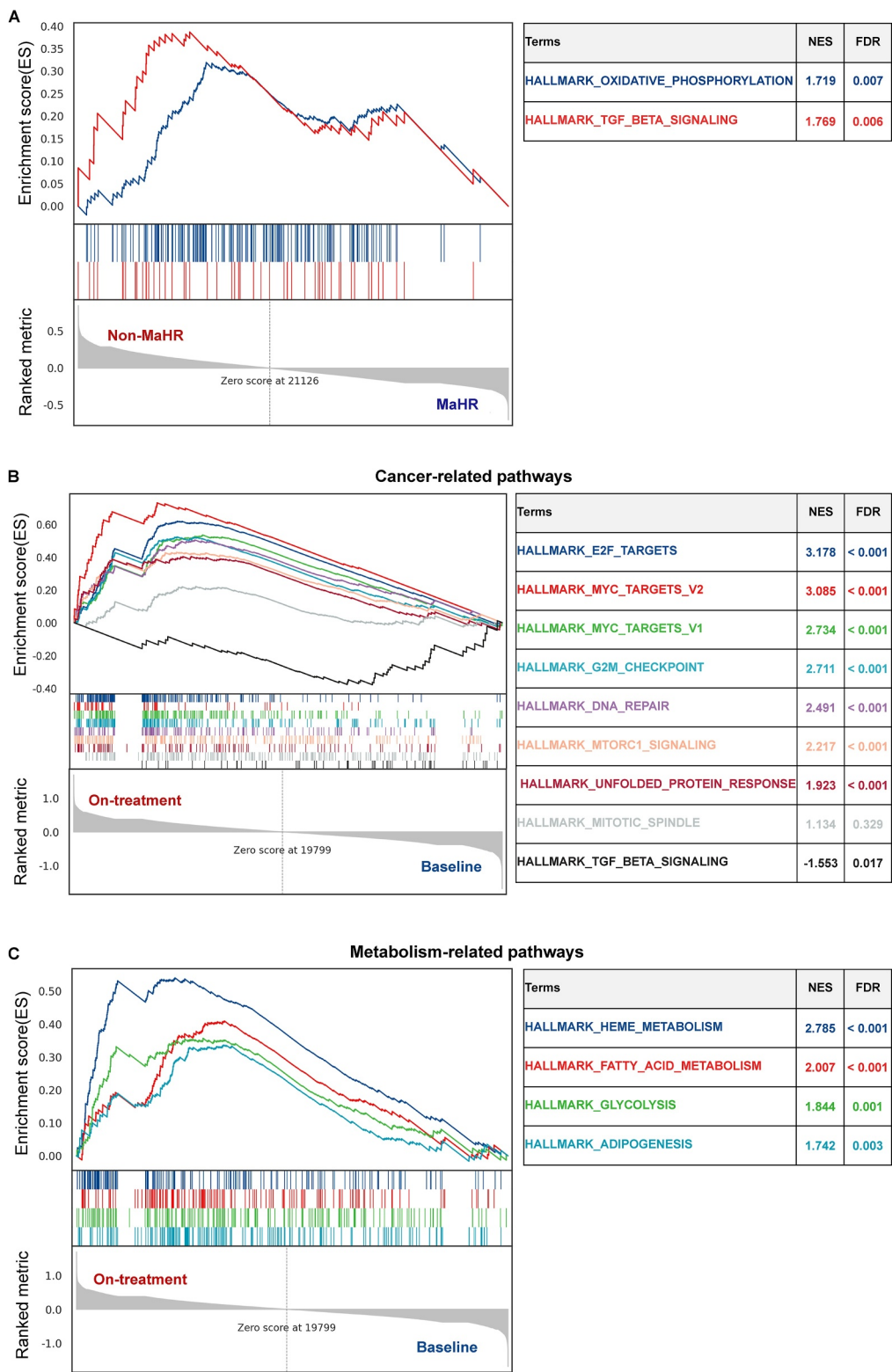


FIGURE 6 Enrichment plots by GSEA and GSVA analyses. (A) Results of the GSEA analysis between patients who did and did not achieve an MaHR at baseline for the gene set of cancer-, metabolism-, oxidative phosphorylation-related pathways. (B–D) Results of the GSEA analysis in patients who did not achieve a MaHR at baseline and on treatment for the gene set of cancer-, metabolism-, oxidative phosphorylation-related pathways. (E) Heatmap of GSVA scores indicating the enrichment levels of cancer-, metabolism-, oxidative phosphorylation-related pathways in all patients, with stratification by both MaHR status and treatment timepoints (baseline and on treatment). (F) Heatmap of average GSVA scores indicating the enrichment levels of cancer-, metabolism-, oxidative phosphorylation-related pathways in all patients, with stratification by both MaHR status and treatment timepoints (baseline and on treatment). (G) Results of the GSEA analysis comparing baseline enrichment of the KRAS signaling pathway between patients who subsequently lost MaHR and those who maintained MaHR during treatment. FDR indicates false-discovery rate; GSEA, gene set enrichment analysis; GSVA, gene set variation analysis; MaHR, major hematologic response; NES, normalized enrichment score.

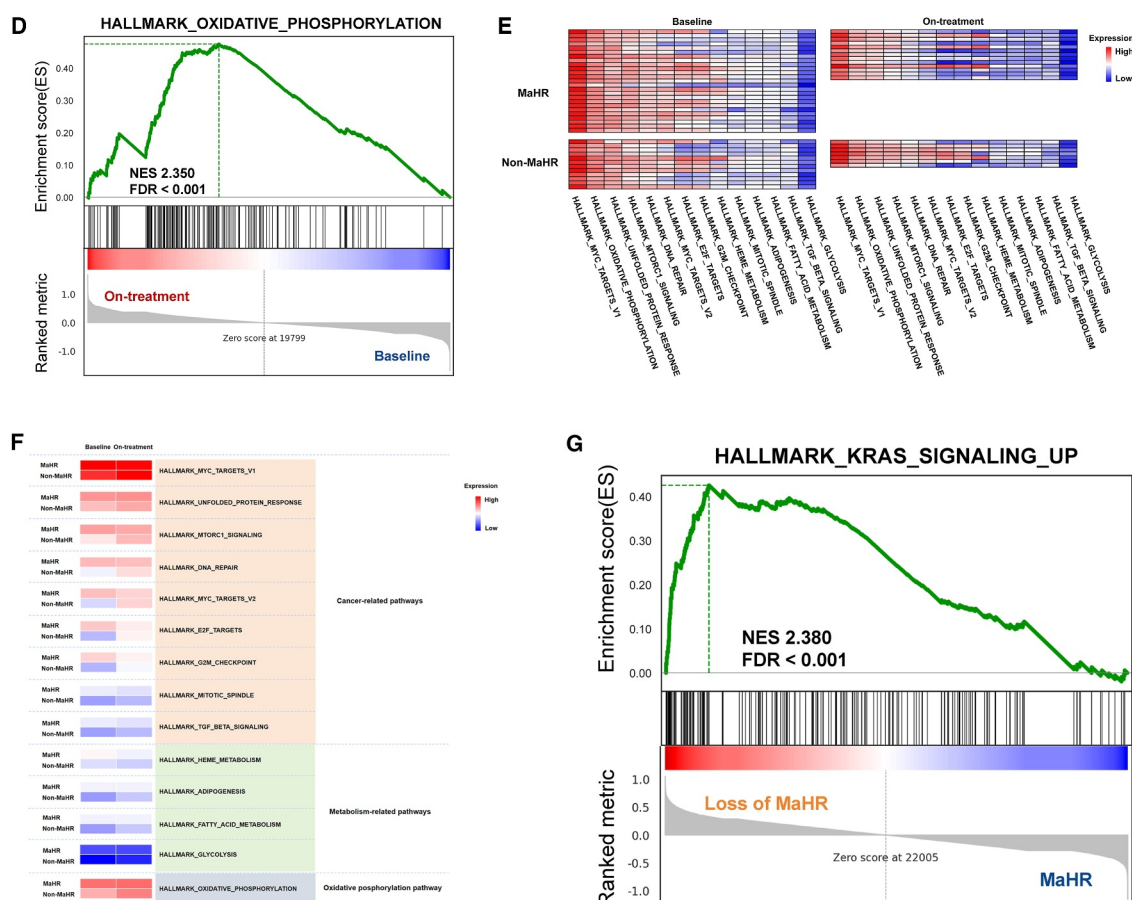


FIGURE 6 (Continued)

metabolism, and heme metabolism), as well as the oxidative phosphorylation pathway, during study treatment compared with baseline (Figure 6B–D). GSEA analyses further corroborated these results (Figure 6E,F).

Next, we focused on baseline data from 25 patients who achieved MaHR. They were divided into two cohorts based on whether they lost MaHR during the follow-up period. GSEA revealed that the 12 patients who lost MaHR during treatment had significant activation of the KRAS signaling pathway compared with those who maintained MaHR ($n = 13$; Figure 6G).

DISCUSSION

We evaluated the efficacy and safety of ponatinib or olverembatinib plus azacitidine in 37 patients with myeloid blast phase CML. Of these heavily pretreated patients, 67% and 81% achieved a MaHR and returned to chronic phase, respectively. Only one patient experienced an early death, which was comparable to other studies.^{7,8,10} Although the median survival was 9 months for all patients, those who underwent transplantation had superior outcomes, with a median survival of 16 months and an encouraging 3-year survival probability of 50%. The treatment responses and outcomes from this

study are consistent with those reported in previous studies combining TKIs with hypomethylating agents.^{7–10,26} Recently, Short et al.¹⁰ reported that 10 of 20 patients (50%) with CML in myeloid blast phase, accelerated phase, or Ph-positive acute myeloid leukemia achieved complete remission or complete remission with incomplete hematologic recovery after two cycles of treatment with ponatinib plus decitabine and venetoclax. The median survival was 11 months in all patients and was not reached in those who underwent transplantation, with a 2-year survival probability of 51%. Indeed, these direct comparisons are challenging because of the small sample size in each study, potential selection biases, variability in transplantation use, and wide CIs. Moreover, previous studies included a certain proportion of patients in lymphoid blast phase or with newly diagnosed blast or accelerated phase, and most patients were 3G-TKI-naïve and underwent transplantation. In contrast, our study mainly included heavily pretreated patients with CML in myeloid blast phase, with 10 of 37 who received prior 3G-TKI exposure, and only one third underwent transplantation.

Some patients failed prior 3G-TKI or TGRX-678 monotherapy but responded after the combination of azacitidine, suggesting that disease progression in some cases may be driven by *BCR::ABL1*-independent mechanisms.²⁷ This is supported by preclinical data indicating that the combination of imatinib or nilotinib plus decitabine

more effectively inhibited the proliferation of K562 cells compared with TKI monotherapy.²⁸

Previous studies on myeloid blast phase CML provided limited data regarding the prognostic significance of genetic abnormalities. In our current study, we observed that *KRAS* and *PTPN11* mutations were significantly associated with worse responses and outcomes. Transcriptomic analyses further corroborated these findings, indicating that patients who did not achieve a MaHR had marked activation of cancer-, metabolism-, and oxidative phosphorylation-related pathways, and the *KRAS* signaling pathway was activated in those who had a brief MaHR. As previous studies reported, mutations in *KRAS* and *PTPN11* activate *RAS* signaling, potentially driving leukemic transformation through increased proliferation and resistance to apoptosis.²⁹ In CML, the *BCR::ABL1* oncoprotein persistently activates Ras-associated pathways, such as the *RAF/MEK/ERK* and *PI3K/AKT/mTOR* pathways. Both in vivo and vitro studies demonstrated that sustained activation of these pathways contributes to TKI resistance in CML.^{14,15,30,31} However, to our knowledge, whether *KRAS* or *PTPN11* mutations affect outcomes has not been investigated in patients with CML in myeloid blast phase. Genomic and transcriptomic analyses in our study revealed that *KRAS* or *PTPN11* mutations and activation of the *KRAS* signaling pathway were significantly correlated with a poor prognosis in patients who had CML in myeloid blast phase and were receiving ponatinib or olverembatinib combined with azacitidine. These findings emphasize the need for future large-scale studies to elucidate the molecular mechanisms and evaluate therapeutic strategies targeting the *KRAS* signaling pathway. In addition, the current findings also indicate that increasing mutation burden was significantly associated with worse PFS, which was understandable.

Our study had several limitations. The sample size was relatively small. However, given the rarity of CML in myeloid blast phase in the TKI era, such limitations are common in this field. Consequently, our point-estimates had large CIs, and the study was underpowered to detect potentially important differences.

In conclusion, we observed that 3G-TKI with azacitidine is an effective and safe therapy for patients who have CML in myeloid blast phase. Moreover, both *RAS* pathway mutations and the marked activation are associated with worse outcomes, which may serve as potential biomarkers for risk stratification and the development of future targeted therapies. Prospective validation in larger cohorts with integrated genomic and transcriptomic profiling is warranted.

AUTHOR CONTRIBUTIONS

Mei Bao: Writing–original draft, formal analysis, data curation, visualization, writing–review and editing, software, and methodology. **Xiao S. Zhang:** Methodology, software, data curation, formal analysis, writing–original draft, writing–review and editing, and visualization. **Zong R. Li:** Data curation. **Lu Yu:** Data curation. **Robert Peter Gale:** Writing–review and editing. **Sha S. Zhao:** Data curation. **Fang Ye:** Data curation. **Cheng C. Yan:** Methodology and software. **Xiao J. Huang:**

Writing–review and editing. **Qian Jiang:** Conceptualization, methodology, investigation, validation, formal analysis, supervision, resources, project administration, visualization, funding acquisition, writing–original draft, writing–review and editing, data curation, and software.

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CONFLICT OF INTEREST STATEMENT

Robert Peter Gale reports personal/consulting fees from Antengene, FFF Enterprises Inc., Hengrui Therapeutic, Janssen Pharmaceuticals Inc., and the Russian Foundation for Basic Research outside the submitted work. Xiaojun Huang reports personal/consulting fees from Ascentage Pharma outside the submitted work. Qian Jiang reports personal/consulting fees from Ascentage Pharma and Novartis Pharma outside the submitted work. The remaining authors disclosed no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data are available upon reasonable request to the corresponding author consistent with the laws of China.

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REFERENCES

- Hochhaus A, Baccarani M, Silver RT, et al. European LeukemiaNet 2020 recommendations for treating chronic myeloid leukemia. *Leukemia*. 2020;34(4):966–984. doi:10.1038/s41375-020-0776-2
- Bonifacio M, Stagno F, Scaffidi L, Krampera M, Di Raimondo F. Management of chronic myeloid leukemia in advanced phase. *Front Oncol*. 2019;9:1132. doi:10.3389/fonc.2019.01132
- Atallah E, Deininger M. Advanced-stage chronic myeloid leukemia: options for difficult treatment situations. *Drugs*. 2025;85(1):41–50. doi:10.1007/s40265-024-02108-2
- Radujkovic A, Dietrich S, Blok HJ, et al. Allogeneic stem cell transplantation for blast crisis chronic myeloid leukemia in the era of tyrosine kinase inhibitors: a retrospective study by the EBMT Chronic Malignancies Working Party. *Biol Blood Marrow Transplant*. 2019;25(10):2008–2016. doi:10.1016/j.bbmt.2019.06.028
- Shah NP, Bhatia R, Altman JK, et al. Chronic Myeloid Leukemia, Version 2.2024, NCCN Clinical Practice Guidelines in Oncology. *J Natl Compr Cancer Netw*. 2024;22(1):43–69. doi:10.6004/jnccn.2024.0007
- Copland M, Slade D, McIlroy G, et al. Ponatinib with fludarabine, cytarabine, idarubicin, and granulocyte colony-stimulating factor chemotherapy for patients with blast-phase chronic myeloid leukaemia (MATCHPOINT): a single-arm, multicentre, phase 1/2 trial. *Lancet Haematol*. 2022;9(2):e121–e132. doi:10.1016/s2352-3026(21)00370-7
- Abaza Y, Kantarjian H, Alwash Y, et al. Phase I/II study of dasatinib in combination with decitabine in patients with accelerated or blast

- phase chronic myeloid leukemia. *Am J Hematol*. 2020;95(11):1288-1295. doi:[10.1002/ajh.25939](https://doi.org/10.1002/ajh.25939)
8. Ruggiu M, Oberkamp F, Ghez D, et al. Azacytidine in combination with tyrosine kinase inhibitors induced durable responses in patients with advanced phase chronic myelogenous leukemia. *Leuk Lymphoma*. 2018;59(7):1659-1665. doi:[10.1080/10428194.2017.1397666](https://doi.org/10.1080/10428194.2017.1397666)
 9. Saxena K, Jabbour E, Issa G, et al. Impact of frontline treatment approach on outcomes of myeloid blast phase CML. *J Hematol Oncol*. 2021;14(1):94. doi:[10.1186/s13045-021-01106-1](https://doi.org/10.1186/s13045-021-01106-1)
 10. Short NJ, Nguyen D, Jabbour E, et al. Decitabine, venetoclax, and ponatinib for advanced phase chronic myeloid leukaemia and Philadelphia chromosome-positive acute myeloid leukaemia: a single-arm, single-centre phase 2 trial. *Lancet Haematol*. 2024;11(11):e839-e849. doi:[10.1016/s2352-3026\(24\)00250-3](https://doi.org/10.1016/s2352-3026(24)00250-3)
 11. et al. Ponatinib and 5-azacytidine for the treatment of chronic myelogenous leukemia in myeloid blast crisis. results of the PONAZA trial (NCT03895671) [abstract]. *HemaSphere*. 2024;8(S1):S170.
 12. Ochi Y. Genetic landscape of chronic myeloid leukemia. *Int J Hematol*. 2023;117(1):30-36. doi:[10.1007/s12185-022-03510-w](https://doi.org/10.1007/s12185-022-03510-w)
 13. Ochi Y, Yoshida K, Huang YJ, et al. Clonal evolution and clinical implications of genetic abnormalities in blastic transformation of chronic myeloid leukaemia. *Nat Commun*. 2021;12(1):2833. doi:[10.1038/s41467-021-23097-w](https://doi.org/10.1038/s41467-021-23097-w)
 14. Agarwal A, Eide CA, Harlow A, et al. An activating KRAS mutation in imatinib-resistant chronic myeloid leukemia. *Leukemia*. 2008;22(12):2269-2272. doi:[10.1038/leu.2008.124](https://doi.org/10.1038/leu.2008.124)
 15. Li X, Pang J, Xue W, et al. Inducible SHP-2 activation confers resistance to imatinib in drug-tolerant chronic myeloid leukemia cells. *Toxicol Appl Pharmacol*. 2018;360:249-256. doi:[10.1016/j.taap.2018.09.044](https://doi.org/10.1016/j.taap.2018.09.044)
 16. Xu L, Chen H, Chen J, et al. The consensus on indications, conditioning regimen, and donor selection of allogeneic hematopoietic cell transplantation for hematological diseases in China—recommendations from the Chinese Society of Hematology. *J Hematol Oncol*. 2018;11(1):33. doi:[10.1186/s13045-018-0564-x](https://doi.org/10.1186/s13045-018-0564-x)
 17. Zhang XH, Chen J, Han MZ, et al. The consensus from the Chinese Society of Hematology on indications, conditioning regimens and donor selection for allogeneic hematopoietic stem cell transplantation: 2021 update. *J Hematol Oncol*. 2021;14(1):145. doi:[10.1186/s13045-021-01159-2](https://doi.org/10.1186/s13045-021-01159-2)
 18. Arber DA, Orazi A, Hasserjian R, et al. The 2016 revision to the World Health Organization classification of myeloid neoplasms and acute leukemia. *Blood*. 2016;127(20):2391-2405. doi:[10.1182/blood-2016-03-643544](https://doi.org/10.1182/blood-2016-03-643544)
 19. Guilhot F, Apperley J, Kim DW, et al. Dasatinib induces significant hematologic and cytogenetic responses in patients with imatinib-resistant or -intolerant chronic myeloid leukemia in accelerated phase. *Blood*. 2007;109(10):4143-4150. doi:[10.1182/blood-2006-09-046839](https://doi.org/10.1182/blood-2006-09-046839)
 20. Wang W, Cortes JE, Tang G, et al. Risk stratification of chromosomal abnormalities in chronic myelogenous leukemia in the era of tyrosine kinase inhibitor therapy. *Blood*. 2016;127(22):2742-2750. doi:[10.1182/blood-2016-01-690230](https://doi.org/10.1182/blood-2016-01-690230)
 21. Cortes J, Apperley J, Lomaia E, et al. Ponatinib dose-ranging study in chronic-phase chronic myeloid leukemia: a randomized, open-label phase 2 clinical trial. *Blood*. 2021;138(21):2042-2050. doi:[10.1182/blood.2021012082](https://doi.org/10.1182/blood.2021012082)
 22. Qin YZ, Jiang Q, Jiang H, et al. Which method better evaluates the molecular response in newly diagnosed chronic phase chronic myeloid leukemia patients with imatinib treatment, BCR-ABL(IS) or log reduction from the baseline level? *Leuk Res*. 2013;37(9):1035-1040. doi:[10.1016/j.leukres.2013.06.003](https://doi.org/10.1016/j.leukres.2013.06.003)
 23. Hänzelmann S, Castelo R, Guinney J. GSEA: gene set variation analysis for microarray and RNA-seq data. *BMC Bioinformatics*. 2013;14(1):7. doi:[10.1186/1471-2105-14-7](https://doi.org/10.1186/1471-2105-14-7)
 24. Subramanian A, Tamayo P, Mootha VK, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102(43):15545-15550. doi:[10.1073/pnas.0506580102](https://doi.org/10.1073/pnas.0506580102)
 25. Sundararajan V, Henderson T, Perry C, Muggivan A, Quan H, Ghali WA. New ICD-10 version of the Charlson comorbidity index predicted in-hospital mortality. *J Clin Epidemiol*. 2004;57(12):1288-1294. doi:[10.1016/j.jclinepi.2004.03.012](https://doi.org/10.1016/j.jclinepi.2004.03.012)
 26. Brioli A, Lomaia E, Fabisch C, et al. Management and outcome of patients with chronic myeloid leukemia in blast phase in the tyrosine kinase inhibitor era—analysis of the European LeukemiaNet Blast Phase Registry. *Leukemia*. 2024;38(5):1072-1080. doi:[10.1038/s41375-024-02204-y](https://doi.org/10.1038/s41375-024-02204-y)
 27. Senapati J, Jabbour E, Kantarjian H, Short NJ. Pathogenesis and management of accelerated and blast phases of chronic myeloid leukemia. *Leukemia*. 2023;37(1):5-17. doi:[10.1038/s41375-022-01736-5](https://doi.org/10.1038/s41375-022-01736-5)
 28. Jiang LC, Luo JM. Role and mechanism of decitabine combined with tyrosine kinase inhibitors in advanced chronic myeloid leukemia cells. *Oncol Lett*. 2017;14(2):1295-1302. doi:[10.3892/ol.2017.6318](https://doi.org/10.3892/ol.2017.6318)
 29. Prior IA, Hood FE, Hartley JL. The frequency of Ras mutations in cancer. *Cancer Res*. 2020;80(14):2969-2974. doi:[10.1158/0008-5472.Can-19-3682](https://doi.org/10.1158/0008-5472.Can-19-3682)
 30. Kaehler M, Osteresch P, Küstner A, et al. Clonal evolution in tyrosine kinase inhibitor-resistance: lessons from in vitro-models. *Front Oncol*. 2023;13:1200897. doi:[10.3389/fonc.2023.1200897](https://doi.org/10.3389/fonc.2023.1200897)
 31. LeMaistre A, Lee MS, Talpaz M, et al. Ras oncogene mutations are rare late stage events in chronic myelogenous leukemia. *Blood*. 1989;73(4):889-891. doi:[10.1182/blood.v73.4.889.889](https://doi.org/10.1182/blood.v73.4.889.889)

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